	Options				Correct Answer
Questions	**A**	**B**	**C**	**D**	
What does garbage collection in Java help to achieve?	Faster execution	Memory management	Increased code complexity	Reduced security	B
What is the extension of compiled Java classes?	.java	.js	.class	.exe	C
Which component of the JDK is used to create documentation for Java code?	javac	javadoc	java	jar	B
Which company originally developed Java?	Microsoft	Sun Microsystems	IBM	Oracle	B
What was the original name of the Java programming language?	Oak	Maple	Pine	Elm	A
Which feature of Java allows it to support multithreading ?	JVM	JDK	Java API	Bytecode	C
How does Java achieve platform independence?	Through the use of the Java compiler	By compiling into native code	By compiling into bytecode	By interpreting code directly	C

What is the role of the Java Runtime Environment (JRE)?	To compile Java code	To provide libraries and JVM	To develop Java applications	To debug Java applications	B
Which command is used to compile Java programs?	java	javac	javadoc	jar	B
What is the primary purpose of the Just-In-Time (JIT) compiler in the JVM?	To compile Java code into bytecode	To convert bytecode into native machine code	To interpret Java code	To garbage collect unused objects	B
What type of garbage collection algorithm is used by default in the HotSpot JVM?	Reference counting	Mark and sweep	Copying collector	Generational garbage collection	D
What is a Java Virtual Machine (JVM) specification?	A set of programming rules	A specification that provides the runtime environment	An IDE for Java development	A debugging tool for Java	B
How can you manually request garbage collection in Java?	System.gc()	Runtime.gc()	Both A and B	Garbage collection cannot be manually requested	C

Which Java feature ensures that code does not violate access restrictions at runtime?	Multithreading	Bytecode verification	Platform independence	Garbage collection	B
What is the role of the javap tool in the JDK?	To compile Java code	To generate API documentation	To disassemble class files	To package Java classes into JAR files	C
How does the JVM handle method overloading?	By using method signatures	By method names	By method return types	By method parameters only	A
Which garbage collection algorithm is used in the G1 (Garbage First) collector in Java?	Mark and sweep	Reference counting	Generational garbage collection	Region-based collection	D
What is the primary difference between a JRE and a JDK?	JRE includes compilers, JDK does not	JDK includes compilers and development tools, JRE does not	JRE includes development tools, JDK does not	JDK is for running Java programs, JRE is for developing them	B
How does the JVM optimize performance using the HotSpot compiler?	By pre-compiling all methods	By interpreting all methods	By selectively compiling methods at runtime	By recompiling all methods in a separate thread	C

What is the size of a byte data type in Java?	4 bits	8 bits	16 bits	32 bits	B
Which keyword is used to create an instance of a class in Java?	create	new	instance	init	B
Which of the following is not a valid Java keyword?	static	Boolean	void	private	B
What is the default value of a boolean variable in Java?	TRUE	FALSE	0	1	B
Which symbol is used for comments in Java?	//	\	**	!!	A
Which data type is used to create a variable that can hold a 64-bit integer?	int	long	short	double	B
Which of the following is not a primitive data type in Java?	int	boolean	String	char	C

Which operator is used to find the remainder in a division operation?	/	%	*	-	B
What is the result of the expression 5 + 2 * 3 in Java?	21	15	11	9	C
Which keyword is used to inherit a class in Java?	inherit	extends	implements	uses	B
Which data type should be used to store a value of 3.14 in Java?	float	int	double	long	C
How many bits are used to represent the char data type in Java?	8	16	32	64	B
Which of the following is a valid declaration of a char in Java?	char ch = 'A';	char ch = "A";	char ch = A;	char ch = A;	A
What will be the output of the following code: int x = 5; System.out. println(x++);	6	5	4	Error	B

What is the correct way to declare a variable in Java?	int 1x=10;	int x=10;	int x=10;	x int=10;	B
Which of the following statements is used to exit from a loop in Java?	exit	stop	break	terminate	C
What is the value of the expression 10 + 20 + "30" + 40 in Java?	100	"10203040"	"30"	"303040"	B
Which keyword is used to prevent a method from being overridden in Java?	final	static	abstract	protected	A
Which of the following correctly defines a constant in Java?	static int MAX=100;	final int MAX=100;	const int MAX=100;	abstract int MAX=100;	B
What is the output of the expression Math.max(1, 4 + 2 * 3)?	7	6	10	4	C

Which of the following is a bitwise operator in Java?	&	&&			
Which data type is used for a variable that stores a single character in Java?	String	char	int	float	B
How do you declare a variable that can hold a value greater than 3.4e38 in Java?	double	float	int	long	A
Which of the following will result in a compile-time error?	int x = (int) 3.9;	float y = 3.14f;	int z = 5.5;	double w = 4;	C
What is the purpose of the break statement in Java?	To skip the current iteration of a loop	To exit a loop or switch statement	To continue to the next iteration of a loop	To return from a method	B
Which of the following is not a looping construct in Java?	for loop	while loop	do-while loop	if-else loop	D

What does the following code print? int x = 5; if (x > 0) { System.out.println("Positive"); } else { System.out.println("Non-positive"); }	Positive	Non-positive	Nothing	Error	A
Which keyword is used to implement decision-making in Java?	switch	case	for	while	A
What is the correct syntax for a while loop in Java?	while condition { }	while (condition) { }	while { condition }	while (condition);	B
Which statement can be used to skip the current iteration of a loop?	break	continue	return	skip	B

What will be the output of the following code? for (int i = 0; i < 5; i++) { System.out.print(i + " "); }	0 1 2 3 4	1 2 3 4 5	0 1 2 3 4 5	1 2 3 4	A
What will be the output of the following code? int x = 10; if (x > 5) { System.out.print("A"); if (x > 8) { System.out.print("B"); } } else { System.out.print("C"); }	A	AB	C	AC	B
Which of the following is true about the switch statement in Java?	It can be used with String variables in Java 7 and later	It can be used with floating-point numbers	It requires a break statement after each case	It can have multiple default cases	A

What is the output of this code? int sum = 0; for (int i = 0; i < 5; i++) { if (i % 2 == 0) continue; sum += i; } System.out.println(sum);	0	4	6	10	B
Which loop would be most appropriate for iterating a known number of times?	while loop	do-while loop	for loop	if-else statement	C
What will be printed by this code? int x = 5; System.out.println(x > 3 ? x < 7 ? "A" : "B" : "C");	A	B	C	AB	A
What is the output of the following code? int i = 0; while (i < 5) { System.out.print(i); i++; if (i == 3) break; }	Ouput:01	Output:012	Output:0123	Output:01234	B

Which statement is true about the do-while loop?	It checks the condition before executing the loop body	It is guaranteed to execute at least once	It cannot be used with the break statement	It must have a counter variable	B
What will be the output of this code? for (int i = 0; i < 3; i++) { for (int j = 0; j < 3; j++) { if (i == j) break; System.out.print(i + "" + j + " "); } }	01 02 10 12 20 21	00 01 02 10 11 12 20 21 22	01 02 12	01 02 10 12 20	D
Which of the following is not a valid loop in Java?	for (;;) { }	while (true) { }	do { } while (true);	loop { } while (true);	D
What is the purpose of the "default" case in a switch statement?	It is executed if no case matches	It must be the last case in the switch	It is optional in a switch statement	Both a and c	D

Consider the following code:
 	
for (int i = 0; i < 3; i++) {
for (int j = 0; j < 3; j++) {
if (i == 1 && j == 1) {
 	
continue;
}
System.out.print(i + "" + j + " ");
}
}
 	
What is the output?
	00 01 02 10 11 12 20 21 22	00 01 02 10 12 20 21 22	00 01 02 10 20 21 22	00 01 02 20 21 22	B

What will be the output of this code?
 	
int x = 5;
switch (x) {
case 1:
System.out.
print("One");
case 5:
System.out.
print("Five");
case 10:
System.out.
print("Ten");
default:
System.out.
print
("Default");
}
	Five	FiveTen	FiveTenDefault	Default	C

What is the result of the following code?
 	
int i = 0;
do {
i++;
if (i == 3)
continue;
System.
out.print(i + "
");
} while (i < 5);
	1 2 3 4 5	1 2 4 5	1 2 4 5 6	1 2 3 4	B

Consider this nested loop:
 	
outer: for (int i = 0; i < 3; i++) {
for (int j = 0; j < 3; j++) {
if (i == 1 && j == 1) {
break outer;
}
System.out.print(i + "" + j + " ");
}
}
 	
What is the output?
	00 01 02 10	00 01 02 10 11	00 01 02 10 20 21 22	00 01 02 10 11 12 20 21 22	A

Which of the following is true about multidimensional arrays in Java?	They must be rectangular (all subarrays must have the same length)	They can be jagged (subarrays can have different lengths)	They are stored in contiguous memory locations	hey can have a maximum of 3 dimensions	B

What is the correct way to iterate over all elements in a 2D array?	for (int i = 0; i < array.length; i++) {
	for (int j = 0; j < array[i].length; j++) {
	// Process array[i][j]
	}
	}
		for (int[] row : array) {
		for (int element : row) {
		// Process element
		}
		}
			for (int i = 0; i < array.length; i++) {
			for (int j = 0; j < array.length; j++) {
			// Process array[i][j]
			}
			}
				Both a and b	D

What is the output of the following code?
int[] numbers = {1, 2, 3, 4, 5};
int sum = 0;
for (int num : numbers) {
sum += num;
if (sum > 6) break;
}
System.out.println(sum);
	6	7	10	15	C

Which of the following statements about arrays in Java is false?	Arrays are objects in Java	The size of an array can be changed after creation	Arrays can store primitive data types directly	Multi-dimensional arrays are implemented as arrays of arrays	B

Consider the following code: int[][][] cube = new int[3][3][]; for (int i = 0; i < cube.length; i++) { for (int j = 0; j < cube[i].length; j++) { cube[i][j] = new int[i + j + 1]; } } System.out.println(cube[2][1].length); What is the output?	3	4	5	Runtime error	B

<table>
<tr>
<td>What is the time complexity of the following method to find a number in a sorted 2D array where each row and column is sorted in ascending order?

public static boolean findNumber(int[][] matrix, int target) {
if (matrix == null || matrix.length == 0 || matrix[0].length == 0) {
return false;
}
int row = 0;
int col = matrix[0].length - 1;
while (row < matrix.length && col >= 0) {
if (matrix[row][col] == target) {
return true;
} else if (matrix[row][col] > target) {
col--;
} else {</td>
<td></td>
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<td></td>
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<td></td>
</tr>
</table>

What is the output of the following code?
 	
int[][] array = {{1, 2}, {3, 4}, {5, 6}};
for (int i = 0; i < array.length; i++) {
for (int j = 0; j < array[i].length; j++) {
if (array[i][j] % 2 == 0) {
 	
continue;
}
System.out.print(array[i][j] + " ");
}
}
	Output: 1 3 5	Output: 2 4 6	Output: 1 2 3 4 5 6	Output: 1 3 5 2 4 6	A

Consider the following method:
 	
public static void rotate(int [][] matrix) {
int n = matrix.length;
for (int i = 0; i < n / 2; i++) {
for (int j = i; j < n - i - 1; j++) {
int temp = matrix[i][j];
matrix[i][j] = matrix[n - 1 - j][i];
matrix[n - 1 - j][i] = matrix[n - 1 - i][n - 1 - j];
matrix[n - 1 - i][n - 1 - j] = matrix[j][n - 1 - i];
matrix[j][n - 1 - i] = temp;
}
}
}
 	
What does this method do?
	Transposes the matrix	Rotates the matrix 90 degrees clockwise	Rotates the matrix 90 degrees counterclockwise	Reverses the matrix	B

What is the output of the following code?
 	
int[][][] arr = new int[3][3][3];
int sum = 0;
for (int i = 0; i < 3; i++) {
for (int j = 0; j < 3; j++) {
for (int k = 0; k < 3; k++) {
arr[i][j][k] = i + j + k;
sum += arr[i][j][k];
}
}
}
System.out.println(sum);
	27	36	54	81	C

Consider the following code snippet:
 	
public static int[] findDiagonalOrder(int[][] matrix) {
if (matrix == null || matrix.length == 0) return new int[0];
int m = matrix.length, n = matrix[0].length;
int[] result = new int[m * n];
int row = 0, col = 0, d = 0;
int[][] dirs = {{-1, 1}, {1, -1}};
for (int i = 0; i < m * n; i++) {
result[i] = matrix[row][col];
row += dirs[d][0];
col += dirs[d][1];
if (row >= m) { row = m - 1; col += 2; d = 1 - d; }
if (col >= n) { col = n - 1; row += 2; d = 1 - d; }
if (row < 0) { row = 0; d					

<table>
<tr><th></th><th></th><th></th><th></th><th></th><th></th></tr>
<tr><td>How can you use bitwise operations to swap two variables `a` and `b` without using a temporary variable?</td><td>`a = a ^ b; b = a ^ b; a = a ^ b;`</td><td>`a = a ^ b; b = b ^ a; a = a ^ b;`</td><td>`a = a & b; b = b & a; a = a & b;`</td><td>`a = a | b; b = b | a; a = a | b;`</td><td>A</td></tr>
<tr><td>What is the result of `12 & ~8` using the bitwise NOT operator in Java?</td><td>12</td><td>8</td><td>4</td><td>0</td><td>C</td></tr>
<tr><td>How do you extract the rightmost set bit from a number `num` using bitwise operators in Java?</td><td>`num & -num`</td><td>`num & ~num`</td><td>`num | -num`</td><td>`num ^ -num`</td><td>A</td></tr>
<tr><td>What is the value of `1 << 3 | 1` in Java?</td><td>8</td><td>9</td><td>5</td><td>7</td><td>B</td></tr>
<tr><td>Which bitwise operator is used to check if a number `num` is a power of two in Java efficiently?</td><td>&</td><td>|</td><td>^</td><td>Operator ==</td><td>A</td></tr>
</table>

<table>
<tr><td>What is the value of `10 >> 1` using the right shift operator in Java?</td><td>5</td><td>2</td><td>1</td><td>0</td><td>A</td></tr>
<tr><td>Which bitwise operation can be used to check if the k-th bit of a number `num` is set in Java?</td><td>`num & (1 << k)`</td><td>`num | (1 << k)`</td><td>`num ^ (1 << k)`</td><td>`num << (1 << k)`</td><td>A</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>How can you clear the lowest set bit of an integer `num` efficiently using bitwise operations in Java?</td><td>`num & (num - 1)`</td><td>`num & ~num`</td><td>`num | -num`</td><td>`num ^ -num`</td><td>A</td></tr>
<tr><td>What is the value of `15 >>> 2` using the bitwise unsigned right shift operator in Java?</td><td>7</td><td>3</td><td>1</td><td>0</td><td>C</td></tr>
</table>

<table>
<tr><td>Which bitwise operator can be used to calculate the absolute difference between two integers `a` and `b` in Java?</td><td>`a & b`</td><td>`a | b`</td><td>`a ^ b`</td><td>`~(a ^ b)`</td><td>D</td></tr>
<tr><td>What is the value of `1 << 31` in Java, considering integer overflow?</td><td>-2147483648</td><td>2147483648</td><td>0</td><td>1</td><td>C</td></tr>
<tr><td>Which bitwise operation can be used to count the number of set bits (1s) in the binary representation of an integer efficiently?</td><td>`num & (num - 1)`</td><td>`num | (num - 1)`</td><td>`num ^ (num - 1)`</td><td>`num << (num - 1)`</td><td>A</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>What is the blueprint for creating objects in Java?</td><td>Object</td><td>Class</td><td>Method</td><td>Constructor</td><td>B</td></tr>
<tr><td>What keyword is used to create an object in Java?</td><td>new</td><td>create</td><td>instance</td><td>init</td><td>A</td></tr>
</table>

Which keyword is used to define a method in a class?	method	def	func	void	D
What is the access modifier for making a class member accessible only within its own class?	public	protected	private	default	C
What is a constructor in Java?	A method to perform calculations	A special method to initialize objects	A method to destroy objects	A method to copy objects	B
What is the purpose of the this keyword in Java?	To refer to the superclass	To refer to the current object	To create new objects	To define static methods	B
How many constructors can a class have in Java?	Only one	Two	One or more	Zero	C
Which keyword is used to define a constant value in Java?	final	static	const	abstract	A
What is the return type of a constructor?	void	int	double	None	D

What does method overloading allow you to do?	Define methods with the same name	Define multiple classes	Create abstract classes	Create static methods	A
Which access modifier allows access within the same package but not from outside the package?	public	private	protected	default	D
What is the correct way to call a constructor of the parent class?	super()	parent()	this()	base()	A
What is the purpose of the static keyword in Java?	To define instance variables	To define class variables	To create objects	To initialize constructors	B
Which of the following is true about constructors in Java?	They can be inherited	They must have a return type	They can be overloaded	They can be abstract	C

What is the output of the following code? public class Test { public static void main (String[] args) { System.out. println(new Test(). hashCode()); } }	A random number	A memory address	Compilation error	0	A
Which keyword is used to inherit a class in Java?	implements	extends	inherits	uses	B
What is the main purpose of a constructor?	To perform I/O operations	To initialize objects	To compare objects	To destroy objects	B
How do you declare a method that can be accessed without creating an instance of the class?	void method()	static void method()	final void method()	abstract void method()	B

What will be the output of the following code? public class Test { int x; public Test() { x = 5; } public static void main(String[] args) { Test t = new Test(); System.out. println(t.x); } }	0	5	Compilation error	Runtime error	B
How can you restrict a method from being overridden?	By declaring it final	By declaring it static	By declaring it abstract	By declaring it private	A
What will be the result of the following code? public class Test { public static void main (String[] args) { System.out. println(new Test(). toString()); } }	The class name and hash code	Only the class name	A runtime error	A compilation error	A
Which of the following is true about this keyword in Java?	It can be used within static methods	It can refer to a class variable	It can refer to the current object instance	It can be used to call a static method	C

What is the significance of the super keyword in Java?	It refers to the superclass	It is used to call a static method	It is used to create an instance	It is used to access private members	A
Which method cannot be overloaded in Java?	main()	static methods	private methods	final methods	A
What will happen if a class has a private constructor?	It cannot be instantiated directly	It can only be inherited	It can only be instantiated once	It can only be instantiated by subclasses	A
What does constructor overloading mean in Java?	Multiple constructors with different signatures	Multiple methods with same name	Overriding constructors from parent class	Using constructor within another constructor	A
Which of the following can access private members of a class?	Subclasses	Methods within the same class	Classes in the same package	Any class in Java	B
How can you call a constructor from another constructor in the same class?	Using super	Using this	Using class name	Using constructor	B

What will be the output of the following code: public static void main(String[] args) { System.out.println (new Test().method()); } in a class Test that has int method() { return 42; }?	0	42	Compile-time error	Runtime error	B
What is the difference between method overloading and method overriding?	Overloading changes return type	Overriding changes return type	Overloading is within same class, overriding involves subclass	Overriding changes parameter list	C
What does the final keyword do when applied to a class?	Allows inheritance	Prevents instantiation	Prevents inheritance	Allows instantiation	C
What is dynamic method dispatch?	Compile-time polymorphism	Run-time polymorphism	Static method call	Dynamic method call	B
Which keyword is used to prevent method overriding?	static	abstract	final	const	C

Can a constructor be inherited?	Yes	No	Only static constructors	Only private constructors	B
What is the main benefit of inheritance?	Improved performance	Code reuse	Better readability	Faster execution	B
What is the type of inheritance where a class is derived from more than one class?	Single inheritance	Multiple inheritance	Multilevel inheritance	Hierarchical inheritance	B
How do you call a parent class constructor from a subclass?	this()	super()	parent()	base()	B
Which of the following statements is true about method overriding?	The method must have a different return type	The method must have the same parameter list	The method must have a different name	The method must be static	B
What is the purpose of the finalize() method in Java?	To clean up resources	To override methods	To prevent inheritance	To stop execution	A
Can a subclass access the private members of its superclass directly?	Yes	No	Only if it's static	Only if it's final	B

What does the instanceof operator do?	Checks the data type	Checks for inheritance	Checks the instance's class	Checks for null	B
Can a class be both abstract and final in Java?	Yes	No	Only in interfaces	Only in enums	B
What happens if a class that inherits another class does not define a constructor?	It cannot be compiled	The default constructor of the superclass is called	It cannot be instantiated	The compiler provides a default constructor	D
What is the purpose of the super keyword when used in a method?	To call a superclass method	To access a local variable	To call a subclass method	To declare a method	A
Which of the following cannot be overridden?	Instance methods	Static methods	Final methods	Constructors	C
Can a subclass override a static method of the superclass?	Yes	No	Only if it's public	Only if it's final	B

What is the result of the following code: public class Test { public static void main(String[] args) { new B().method(); } } class A { void method() { System.out.println("A"); } } class B extends A { void method() { System.out.println("B"); } }	A	B	Compile-time error	Runtime error	B
What is covariant return type in Java?	Method returns superclass type	Method returns subclass type	Method returns the same type	Method returns a different type	B
Can a final method be inherited in Java?	Yes	No	Only if it's static	Only if it's private	A
What happens if a superclass and subclass have a static method with the same name and signature?	Compile-time error	The superclass method is called	The subclass method hides the superclass method	Runtime error	C

Can the super keyword be used to call a superclass method in the constructor?	Yes	No	Only if it’s static	Only if it’s private	A
Which of the following is true about the finalize method?	It is called by the garbage collector	It is called manually	It can be overridden	It cannot be overridden	A
What is the main purpose of dynamic method dispatch?	Memory efficiency	Code readability	Runtime polymorphism	Compile-time polymorphism	C
What will be the output of the following code: class A { final void method() { System.out. println("A"); } } class B extends A { void method() { System.out. println("B"); } public static void main (String[] args) { new B(). method(); } }	A	B	Compile-time error	Runtime error	C